Q $y = \arccos(x)$ $\frac{dy}{dx}$?

A $\arccos(x) = y$

$\cos(y) = x$

$\frac{d}{dx}$ ↓

$\frac{d}{dx}(\cos(y)) = \frac{d}{dx}(x)$

$-\sin(y) \cdot \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{-1}{\sin(y)}$ ← Right side can't have y.

Side $\sin^2 y + \cos^2 y = 1$

$\sin^2 y = 1 - \cos^2 y$

$\sin y = \pm\sqrt{1-\cos^2 y}$

$\sin y = \sqrt{1-\cos^2 y}$

$\frac{dy}{dx} = \frac{-1}{\sqrt{1-\cos^2 y}}$

$\frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}}$